

Learning4help by Waqas Qureshi

Assalam-o-Alaikum! In this file we'll provide you **CS301P Important Midterm MCQs. Must Prepare before Midterm Exams.**
Verify Answers yourself too. I am not responsible for any answer. If you found mistake then inform me.
If you found mistake then let us know.

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CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Time Left sec(s)

Question # 1 of 10 (Start time: 11:36:55 PM, 26 November 2024) Total Marks: 1

Binary search tree class is defined as a template class so that it can be used for _____ datatype(s).

Select the correct option

Reload Math Equations

☐ Strings

☐ Integer

☒ Any

☐ Character

Search

CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 2 of 10 (Start time: 11:37:46 PM, 26 November 2024) Total Marks: 1

In recursive functions, recursion is stopped by using:

Select the correct option

Reload Math Equations

☒ Terminating conditions

☐ Function Signature

☐ Binary Search trees

☐ Function prototype

Search

CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 3 of 10 (Start time: 11:38:15 PM, 26 November 2024) Total Marks: 1

How many comparison(s) will be made at each level of a BST for searching a number which is not present in BST?

Select the correct option

☐ 2

☐ 3

☒ 1

☐ 4

Reload Math Equations

What does the following traversal code do?

```
void func (TreeNode* treeNode) {  
    if( treeNode != NULL ) {  
        func(treeNode->getLeft());  
        func(treeNode->getRight());  
        cout << *(treeNode->getInfo())<<" ";  
    }  
}
```

Select the correct option

☒ Post-order traversal

☐ Preorder traversal

☐ Level-order traversal

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CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 5 of 10 (Start time: 11:39:12 PM, 26 November 2024) Total Marks: |

Which of the following memory portion is used to dynamically allocate the memory?

Select the correct option

Stack

Code

Queue

Heap

Reload Math Equations

CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 6 of 10 (Start time: 11:39:45 PM, 26 November 2024) Total Marks: |

The code of BST traversal methods becomes very short by using:

Select the correct option

Heap

Recursive calls

Non-recursive calls

Friend functions

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CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 7 of 10 (Start time: 11:40:21 PM, 26 November 2024) Total Marks: 1

Which of the following represents the destructor of Binary Search Tree class?

Select the correct option

☒ ~BinarySearchTree()

☐ void ~BinarySearchTree()

☐ BinarySearchTree() ~

☐ BinarySearchTree()

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CS301P - Data Structures (Practical) (Graded Quiz - 2) Quiz Start Time: 11:36 PM, 26 November 2024

Question # 8 of 10 (Start time: 11:41:01 PM, 26 November 2024) Total Marks: 1

A binary tree in which every non-leaf node has non-empty right and left subtrees is called

Select the correct option

☒ Strictly Binary Tree

☐ AVL Tree

☐ Huffman Tree

☐ Graph

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Which of the following traversal method will be used, if we want the output as 3,4,7,9,14,15,16,18,20?

```
graph TD; 14((14)) --- 4((4)); 14 --- 15((15)); 4 --- 3((3)); 4 --- 9((9)); 9 --- 7((7)); 15 --- 18((18)); 18 --- 16((16)); 18 --- 20((20));
```

Select the correct option

☐ Preorder traversal

☐ Level-order traversal

☐ Post-order traversal

☒ In-order traversal

Reload Math Equations

CS301P - Data Structures (Practical) (Graded Quiz - 2)

Quiz Start Time: 11:36 PM, 26 November 2024

Question # 10 of 10 (Start time: 11:42:16 PM, 26 November 2024) Total Marks: 1

In C++, ampersand '&' symbol is used for:

Select the correct option

☒ Reference Variables

☐ Arrays

☐ Static variables

☐ Pointers

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CS301P - Data Structures (Practical) (Attendance Based Quiz - 7) Quiz Start Time: 10:41 PM, 29 November 2024

Question # 2 of 5 (Start time: 10:42:16 PM, 29 November 2024) Total Marks: 1

If you create a BST with data that is sorted in an ascending or descending order, it will be similar to:

Select the correct option

AVL Tree

Linked List

Complete Binary Tree

Strictly Binary Tree

Reload Math Equations

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CS301P - Data Structures (Practical) (Attendance Based Quiz - 7) Quiz Start Time: 10:41 PM, 29 November 2024

Question # 3 of 5 (Start time: 10:42:57 PM, 29 November 2024) Total Marks: 1

Which of the following function prototype is declaring the function as constant?

Select the correct option

const EType& findMin() const;

const EType& findMin();

EType& findMin(const int a);

void EType& findMin(const int a);

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CS301P - Data Structures (Practical) (Attendance Based Quiz - 7) Quiz Start Time: 10:41 PM, 29 November 2024

Question # 4 of 5 (Start time: 10:43:43 PM, 29 November 2024) Total Marks: 1

A tree will be an AVL tree if _____ of the tree fulfills the AVL conditions.

Select the correct option

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☒ Every node

☐ Leaf nodes

☐ Non-leaf nodes

☐ Root node

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CS301P - Data Structures (Practical) (Attendance Based Quiz - 7) Quiz Start Time: 10:41 PM, 29 November 2024

Question # 5 of 5 (Start time: 10:44:10 PM, 29 November 2024) Total Marks: 1

What will be the height of an empty AVL tree?

Select the correct option

Reload Math Equations

☐ 1

☒ -1

☐ 2

☐ 0

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Time Left: sec(s)

CS301P - Data Structures (Practical) (Regular Quiz 2)

Quiz Start Time: 03:06 PM, 22 May 2024

Question # 1 of 10 (Start time: 03:06:55 PM, 22 May 2024) Total Marks: 1

Recursion is performed with the help of _____ data structure.

Select the correct option

☐ Binary Tree

☒ Stack

☐ Linked List

☐ Queue

Reload Math Equations

Which of the following traversal method will be used, if we want the output as 9,4,7,9,14,15,16,18,20?

```
graph TD; 14((14)) --- 4((4)); 14 --- 15((15)); 4 --- 3((3)); 4 --- 9((9)); 9 --- 7((7)); 15 --- 18((18)); 18 --- 16((16)); 18 --- 20((20));
```

Select the correct option

☐ Post-order traversal

☒ In-order traversal

Reload Math Equations

Question # 3 of 10 (Start time: 03:08:49 PM, 22 May 2024) Total Marks: 1

The _____ level of any leaf in a binary tree is called the depth of the tree

Select the correct option

☐ Minimum

☒ Zero

☐ Maximum

☐ Mid

Click to Save Answer & Move to Next Question

Question # 4 of 10 (Start time: 03:10:22 PM, 22 May 2024) Total Marks: 1

Which of the following queue does not always follow FIFO behavior?

Select the correct option

☐ Circular queue

☐ Simple queue

☒ Priority queue

☐ Double-ended queue

Click to Save Answer & Move to Next Question

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CS301P - Data Structures (Practical) (Regular Quiz 2) Quiz Start Time: 03:06 PM, 22 May 2024

Question # 5 of 10 (Start time: 03:12:12 PM, 22 May 2024) Total Marks: 1

Which of the following symbol is used for assigning addresses to pointer variables?

Select the correct option

☒ *

☐ +

☐ ~>

☐ >

☐ >

CS301P - Data Structures (Practical) (Regular Quiz 2) Quiz Start Time: 03:06 PM, 22 May 2024

Question # 6 of 10 (Start time: 03:12:48 PM, 22 May 2024) Total Marks: 1

Which of the following represents the destructor of Binary Search Tree class?

Select the correct option

☐ BinarySearchTree()

☐ void ~BinarySearchTree()

☒ ~BinarySearchTree()

☐ BinarySearchTree() ~

Question # 7 of 10 (Start time: 03:13:29 PM, 22 May 2024) Total Marks: 1

In a binary tree, level of root node is :

Select the correct option

☒ 0

☐ Same as the level of its right child

☐ Same as the level of its left child

☐ 2

Click to Save Answer & Move to Next Question

Question # 8 of 10 (Start time: 03:16:00 PM, 22 May 2024) Total Marks: 1

The code of BST traversal methods becomes very short by using:

Select the correct option

☒ Non-recursive calls

☐ Heap

☐ Recursive calls

☐ Friend functions

Click to Save Answer & Move to Next Question

Question # 9 of 10 (Start time: 03:17:36 PM, 22 May 2024) Total Marks: 1

The following method getLeft () of TreeNode class returns the _____ to the left sub-tree.

```
TreeNode * getLeft(){  
    return left;  
}
```

Select the correct option

☐ Object

☒ Pointer

☐ None of the given options

☐ Integer value

Reload Math Equations

Question # 10 of 10 (Start time: 03:19:05 PM, 22 May 2024) Total Marks: 1

What does the following traversal code do?

```
void func (TreeNode* treeNode) {  
    if( treeNode != NULL ) {  
        func(treeNode->getLeft());  
        func(treeNode->getRight());  
        cout << *(treeNode->getInfo())<<" ";  
    }  
}
```

Select the correct option

☒ Preorder traversal

☐ Post-order traversal

Reload Math Equations